Kieran Huynh

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Australian citizen · Available full-time Nov 2026 to Feb 2027 (summer break), part-time during semester

PROFILE

2nd-year Mechanical Engineering (Honours) and Commerce student (WAM 83) seeking a mechanical or mechatronic design internship. Project work spans a Metso industry design challenge (3rd of 60+ teams), a self-funded 5-DOF robotic arm, and a camera housing field-tested at Narrabeen beach. CAD (SolidWorks, Fusion 360, Onshape), FEA, 3D printing and design for CNC machining.

ENGINEERING PROJECTS

Turbimax Sensor Mounting Arm · Metso designathon, team of 4, across 4 days *2026*

- Owned concept, CAD, engineering drawings and BOM for a telescoping arm that positions an Endress+Hauser Turbimax CUS50D sensor and CYA112 immersion tube on a thickener bridge. Placed 3rd of 60+ teams, judged on design and feasibility.
- Rejected folding arms (poor reach control) and a fixed sliding bar (would obstruct the maintenance walkway). Chose telescoping RHS beams sized from BlueScope tables, with a pivot joint and spring-loaded detents; structural and material analysis by teammates.

5 DOF Robotic Arm · Self-funded *In progress (target Feb 2027)*

- Building an arm from 3D-printed and salvaged parts (Onshape, Arduino, MG995 / MG90S servos, sized from moment calculations) to return loose items to their places and clear finished prints, so the workshop can self-clean while unattended.
- Reach target met; joint speed selectable at 40, 30 or 20°/s. Diagnosed a 5 A supply cut-out caused by simultaneous servo inrush at power-up. 200 g payload test at 40 cm pending the gripper; MATLAB inverse kinematics next.

Jacking Bar Bracket · USYD Motorsport (FSAE) induction task *2025*

- Redesigned in 6061 aluminium for a team-specified 700 N lifting load: 98.2 g vs 178 g for last year's mild steel part (45% lighter), FOS 4.47 (SolidWorks Simulation, mesh convergence and fastener checks), with a CAM toolpath.

Seagull: AI Swimmer-Detection Housing · HSC major work (solo) *2024*

- Designed a pole-mounted housing for a YOLO swimmer-detection system (Raspberry Pi 4 with AI camera), prioritising weight and size reduction: 15 Fusion 360 iterations from plywood fit check to 3D-printed shell. System field-tested for an hour at Narrabeen. 2nd in NSW HSC Design and Technology.

Pneumatic Engine · USYD engineering unit, team of 6

- Led the team and designed the engine. Re-machined the piston head after out-of-tolerance parts caused air loss; ran for 60 s at 7 PSI.

ENGINEERING EXPERIENCE

Maverick Program Member, Vexev · Vascular imaging company *Oct 2026 – Present*

- Selected through technical interviews and a design assessment (an inclined motor-rail test jig for 60°C) for a vetted pool of student and graduate engineers.

Work Experience, Tricycle Developments · Industrial design and engineering group *Nov 2023 (2 weeks)*

- Produced SolidWorks CAD of the housing for a childhood epilepsy monitoring device in a six-person team.

LEADERSHIP AND WORK EXPERIENCE

External Affairs Director (2026) / Sponsorship Officer (2025) · Sydney Uni Engineering Undergraduate Assoc. *2025 – Oct 2026*

- Grew sponsorship revenue from $2,400 to $10,650 and sponsors from 2 to 14, leading a team of 5.
- Ran 16 industry speaker events and 3 networking nights (about 250 attendees each) across 21 organisations, including ResMed, Dassault Systèmes and Engineers Australia.

Finance Administrator, Lonsec Research *Oct 2025 – Present · 2 days/week*

- Proposed a $40,000 Phase 1 systems integration project (Salesforce, Workday, iRate) after the MYOB-to-Workday migration; approved by the CTO and CFO and delivered with software engineers.
- Automated overdue-invoice chasing; managed accounts receivable and payable and prepared audit evidence.

Office Manager, then Finance, KBH Migration *Nov 2024 – Oct 2025*

- Prepared 150+ contracts, ran Xero accounts and payroll, and set up health-insurance partnerships that created a new commission revenue stream.

EDUCATION

Bachelor of Mechanical Engineering (Honours) and Commerce · University of Sydney *Feb 2025 – Expected 2030*

- Distinction average (WAM 83) · Dalyell Scholar (USYD advanced scholars program)

HSC · Sydney Grammar School *2024*

- ATAR 98.9 · 2nd in NSW Design and Technology · School D&T Prize (2023, 2024)

SKILLS

CAD and analysis: SolidWorks, Fusion 360, Onshape · SolidWorks Simulation (FEA) · engineering drawings

Making: FDM and resin 3D printing, laser cutting · design for CNC machining and CAM toolpaths

Programming and electronics: Arduino, Raspberry Pi, MATLAB, Python

Business tools: Excel, Xero, Workday, Salesforce, Jira